

Robel Kebede Gebre, PhD

Assistant Professor of Radiology, Mayo Clinic

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Professional Profile

I develop PET and MRI biomarkers that characterize heterogeneity and progression in neurodegenerative disease. My research combines medical imaging, machine learning, explainable artificial intelligence, and disease progression modeling, with current emphasis on Alzheimer disease and related dementias, tau PET quantification, MRI harmonization, cerebrovascular disease, and imaging differences between multiple system atrophy and Parkinson disease.

Appointments and Research Experience

Mayo Clinic, Department of Radiology, Rochester, Minnesota

2021–present

- **Current appointment:** Assistant Professor of Radiology.
- **Previous appointment:** Postdoctoral Research Fellow / Research Associate.
- **Research areas:** PET and MRI biomarker development; tau PET quantification; scanner harmonization; multimodal cognition prediction; Alzheimer disease genetics; brain age modeling; cerebral small vessel disease and cerebral amyloid angiopathy; multiple system atrophy and Parkinson disease; clinical trial enrichment and anti-amyloid treatment eligibility.
- **Methodological contributions:** THETA tau PET summary measure, Transition Point Estimator (TPE), PET tracer classification from image appearance, imaging harmonization methods, machine learning models for disease heterogeneity and progression, and SNP interaction encoding methods.

Education

Doctor of Philosophy in Medicine (Medical Physics and Technology), University of Oulu, Finland

2021

Master of Science in Biomedical Engineering and Clinical Technology, Czech Technical University, Czech Republic and University of Groningen, The Netherlands

2016

Bachelor of Science in Biomedical Engineering, Jimma University Institute of Technology, Ethiopia, *Summa Cum Laude*

2013

Awards and Honors

- Don Summers Memorial MSA Travel Fellowship Award (2025)
- Alavi-Mandell Award (2025)
- Young Investigator Travel Scholarship, Human Amyloid Imaging (2023)
- Mayo Clinic Radiology Research Fellowship, Mayo Clinic (2021)
- PhD work featured by *Tekniikka&Talous* and *Kauppalehti*, Finland (2021)
- Orthopedic Research Society Travel Scholarship Award (2019)
- Common European Master's Course in Biomedical Engineering Scholarship, Erasmus Mundus (2013)
- Summa Cum Laude, Jimma University (2013)
- Named best young scientist in Ethiopia for BSc project, "Wireless Infant Incubators"

Invited and Selected Presentations

- Alzheimer's Association International Conference (AAIC), 2026
- University of Mississippi Medical Center, MIND Center, 2025
- Karolinska Institutet, Department of Neurobiology, Sweden, 2024
- Mayo-Karolinska Institutet Seminar Series, 2022

Publications and Peer-Reviewed Conference Papers

1. **Gebre RK**, Reid RI, Senjem ML, et al. Automated PET tracer classification in multi-site, multi-cohort studies using deep learning. *EJNMMI Research*. 2026. DOI.

2. **Gebre RK**, Raghavan S, De Tora MEJ, et al. Precise disease heterogeneity and progression quantification in MSA and Parkinson's disease using machine learning. *Scientific Reports*. 2026;16:10579. [DOI](#).
3. Raghavan S, Przybelski SA, **Gebre RK**, et al. Regional diffusion imaging measures to disentangle SVD-related hypertensive arteriopathy versus cerebral amyloid angiopathy. *Molecular Neurodegeneration*. 2026. [DOI](#).
4. Raghavan S, **Gebre RK**, Vemuri P. Artificial intelligence: relevance and applications in dementia. In: Sharma CP, Thomas V, Nakhmani A, Smith RJ, eds. *Artificial Intelligence and Brain-Computer Interfaces in Healthcare*. Academic Press; 2026:143–153.
5. **Gebre RK**, Przybelski SA, Raghavan S, et al. Precise estimation of heterogenous global tau burden in the brain sheds light on cognitive resilience mechanisms. *Alzheimer's & Dementia*. 2025;21(S2):e103992. [DOI](#).
6. Reid RI, **Gebre RK**, Kamykowski MG, et al. Automatic classification of head MR series. *Alzheimer's & Dementia*. 2025;21(S8):e109882. [DOI](#).
7. Reyes CD, **Gebre RK**, Gunter JL, et al. Harmonization remains unresolved: the case of white matter hyperintensity using ComBat and Deming regression. *Alzheimer's & Dementia*. 2025;21(S7):e108630. [DOI](#).
8. Satpathi S, **Gebre RK**, Gunter JL, et al. Developing scanner change invariant brain age models for aging and dementia studies. *Alzheimer's & Dementia*. 2025;21(S2):e097891. [DOI](#).
9. Raghavan S, Przybelski SA, **Gebre RK**, et al. Diffusion imaging measures to disentangle SVD-related hypertensive arteriopathy versus cerebral amyloid angiopathy. *Cerebral Circulation - Cognition and Behavior*. 2025;9:100430. [DOI](#).
10. **Gebre RK**, et al. Advancing Tau PET quantification in Alzheimer disease with machine learning: introducing THETA, a novel tau summary measure. *Journal of Nuclear Medicine*. 2024. [DOI](#).
11. **Gebre RK**, et al. Can integration of Alzheimer's plasma biomarkers with MRI, cardiovascular, genetics, and lifestyle measures improve cognition prediction? *Brain Communications*. 2024. [DOI](#).
12. Gunter NB, **Gebre RK**, et al. Machine learning models of polygenic risk for enhanced prediction of Alzheimer disease endophenotypes. *Neurology: Genetics*. 2024. [DOI](#).
13. Raghavan S, Przybelski SA, Lesnick TG, et al., **Gebre RK**, et al. Vascular risk, gait, behavioral, and plasma indicators of VCID. *Alzheimer's & Dementia*. 2024; [DOI](#).
14. Carvalho D, Gunter N, **Gebre RK**, et al. Mapping neuroimaging using artificial intelligence to detect hypersomnia and its neurobiological correlates. *Sleep Medicine*. 2024;115:S76–S77. [DOI](#).
15. **Gebre RK**, Jack CR, Vemuri P. Identifying transition points in AD biomarkers using machine learning: a supplement to the reliable worsening method. *Alzheimer's & Dementia*. 2024;20(S9):e093902. [DOI](#).
16. **Gebre RK**, Moscoso A, Raghavan S, et al. Validation of the Tau Heterogeneity Evaluation in Alzheimer's Disease (THETA) score using longitudinal and histopathology data. *Alzheimer's & Dementia*. 2024;20(S1):e091770. [DOI](#).
17. Yu X, Przybelski SA, Reid RI, **Gebre RK**, et al. Chronic medical conditions and dementia risk: brain age models for quantifying impact and understanding mechanisms. *Alzheimer's & Dementia*. 2024;20(S7):e089382. [DOI](#).
18. **Gebre RK**, et al. Cross-scanner harmonization methods for structural MRI may need further work: a comparison study. *NeuroImage*. 2023; [DOI](#).
19. Ramanan VK, **Gebre RK**, et al. Genetic risk scores enhance the diagnostic value of plasma biomarkers of brain amyloidosis. *Brain*. 2023. [DOI](#).
20. Tian J, Raghavan S, Reid RI, et al., **Gebre RK**, et al. White matter degeneration pathways associated with tau deposition in Alzheimer disease. *Neurology*. 2023. [DOI](#).
21. Raghavan S, Przybelski SA, Lesnick TG, et al., **Gebre RK**, et al. Non-imaging measures to predict variability in white matter changes relevant to VCID. *Alzheimer's & Dementia*. 2023;19(S17):e077200. [DOI](#).
22. Yu X, **Gebre RK**, Przybelski SA, et al. NODDI may be more sensitive to neurodegenerative changes than structural MRI. *Alzheimer's & Dementia*. 2023;19(S17):e076509. [DOI](#).
23. **Gebre RK**, et al. Detecting hip osteoarthritis on clinical CT: a deep learning application based on 2-D summation images derived from CT. *Osteoporosis International*. 2022. [DOI](#).
24. **Gebre RK**, et al. Discrimination of low-energy acetabular fractures from controls using computed tomography-based bone characteristics. *Annals of Biomedical Engineering*. 2021. [DOI](#).
25. Bhuyan M, **Gebre RK**, Finnäll M, et al. Preparation of filter by alkali activation of blast furnace slag and its application for dye removal. *Journal of Environmental Chemical Engineering*. 2021;10:107051. [DOI](#).
26. **Gebre RK**, et al. Structural risk factors for low-energy acetabular fractures. *Bone*. 2019. [DOI](#).

Research Grants

Active

- Investigating Resistance and Resilience Mechanisms in Alzheimer's Disease. National Institute on Aging (FP00090851.02-PR08) - Collaborator.

Completed

- Radiology Research Fellowship, Mayo Clinic Radiology - Recipient.

Methods and Software Developed

- **THETA**: Tau Heterogeneity Evaluation in Alzheimer's Disease, a machine learning tau PET summary measure. [GitHub](#)
- **Transition Point Estimator (TPE)**: Machine learning approach for identifying biomarker transition points. [GitHub](#)
- **Two-Way Epistatic Interaction Encoding**: Approach to encode pairwise SNP interactions. [arXiv:2306.09175](#)
- **PET Tracer Classification**: Deep learning framework for identifying PET tracer identity directly from image appearance across multiple cohorts.

Peer Review Service

Alzheimer's & Dementia; Applied Sciences; Artificial Intelligence in Medicine; BMC Neurology; Bone; Bone Reports; Brain Communications; Brain Sciences; Diagnostics; European Journal of Neuroscience; Human Brain Mapping; IEEE Journal of Translational Engineering in Health and Medicine; Information; Insights into Imaging; Life; Nature; Nature Communications; Osteoarthritis and Cartilage; PLOS ONE; Scientific Reports; The Egyptian Journal of Neurology, Psychiatry and Neurosurgery.

Professional Memberships

American Autonomic Society (AAS); Alzheimer's Association International Society to Advance Alzheimer's Research and Treatment (ISTAART).

Technical Certification

Philips Healthcare Systems - MRI Field Service Engineer, Achieva 1.5T (2016).

Languages

Amharic; English.